



Hashing, Homework 5 - Solution

1 Universal Hash Family.

1. Let $U = \{1, 2, 3, 4\}$, and consider a table at size $m = 2$. Find 4 functions $h_1, h_2, h_3, h_4 : U \rightarrow [m]$ such that the family $\mathcal{H} = \{h_1, h_2, h_3, h_4\}$ is universal.
2. Give an example of a family of functions $\mathcal{H} = \{h : U \rightarrow [m]\}$ such that for any $x \in U$ and $j \in [m]$, on the one hand:

$$\Pr[h(x) = j] \leq \frac{1}{m}$$

Yet, on the other hand, $\mathcal{H}$ is not universal.

3. U is the set of all matrices $\mathbb{F}_2^{n \times n}$ and $h_X : U \rightarrow \{0, 1\}$, $h_X(A) = \text{Tr}(XA)$. Prove that $\mathcal{H} = \{h_X : X \in \mathbb{F}_2^{m \times n}\}$ is a universal family.

Solution.

1. Consider, the following suggestion:

$$\begin{array}{c} 1 \ 2 \ 3 \ 4 \\ h_1 = 1 \ 1 \ 2 \ 1 \\ h_2 = 1 \ 1 \ 2 \ 2 \\ h_3 = 1 \ 2 \ 1 \ 1 \\ h_4 = 1 \ 2 \ 1 \ 2 \end{array}$$

Any two different keys $x \neq y \in \{1, 2, 3, 4\}$ have at most two functions, out of four, that cause a collision between them. Therefore, when sampling h uniformly from $\mathcal{H}$ we have:

$$\Pr[h(x) = h(y)] \leq \frac{2}{4} = \frac{1}{2} = \frac{1}{m}$$

2. Consider the constant functions family, $\mathcal{H}_c = \{h_j : h_j(x) = j \forall x, j \in [m]\}$. Thus for any j :

$$\Pr[h(x) = j] = \Pr[h_j \text{ was picked}] = \frac{1}{m}$$

Yet, since all the functions in $\mathcal{H}_c$ are constant functions, then for any $x \neq y \in U$, and any $h \in \mathcal{H}$, we have the equality $h(x) = h(y)$. So

$$\Pr[h(x) = h(y)] = 1 > \frac{1}{m}$$

3. By the definition of trace:

$$h_X(A) = \mathbf{Tr}(XA) = \sum_i^n (XA)_{ii} = \sum_i^n X_{ij}A_{ji}$$

So, the hashing is equivalence to flattening A into an $n \times n$ binary vector and then multiplying it by a randomly chosen vector. Which was proved in the recitation to be a universal.

2 Universal Hashing

1. Prove by yourself the following claim.¹

Claim 1. Let $\mathcal{H} = \{h : U \rightarrow [m]\}$ be a universal hash family. Denote by T an m -cell table, and suppose we hash n items $x_1, \dots, x_n$ into T . Then, for any x_i , the expected number of items that are hashed to the same cell as x_i is:

$$1 + \frac{n-1}{m}$$

2. Let U be a set of keys. Prove the following claim.

Claim 2. When $|U| \geq m \cdot n$, for every hash table of size m that uses the chaining method, there exists a subset $S \subset U$ of size n such that every element in S is mapped to the same table cell.

Solution.

1. We use the notation that was given in the recitation. For any $x, y \in I$, let $X_{x,y}$ be the indicator of the event $h(x) = h(y)$. Observe that $C_x = \sum_{y \in I} X_{x,y}$. Therefore, by the linearity of expectation and $\mathcal{H}$ being a universal hash family, we have:

$$\begin{aligned} \mathbf{E}[C_x] &= \mathbf{E}\left[\sum_{y \in I} X_{x,y}\right] = \sum_{y \in I} \mathbf{E}[X_{x,y}] = \sum_{y=x \in I} \mathbf{E}[X_{x,y}] + \sum_{y \neq x \in I} \mathbf{E}[X_{x,y}] \\ &= 1 + \sum_{y \neq x \in I} \Pr[h(x) = h(y)] \leq 1 + \frac{n-1}{m} \end{aligned}$$

¹Given and proven in both the lecture and the recitation.

2. Assume that $|U| = m \cdot n$. So we can decompose U into a disjoint union of m subsets $S_1, \dots, S_m$. Notice that for every $i \in [m]$, h maps fewer than n of the elements in the union of S_j 's subsets to the i th cell. Otherwise, since they are disjoint, the combination of those elements together yields a subset $S \subset U$ of size n which h maps all its elements to i .

On the other hand, we can count the elements of U by counting the assignments induced by the application of h on $U = \cup S_j$. Each cell in $[m]$ counts equal to or less than $n - 1$ elements that map to it, so in overall:

$$|U| \leq m \cdot (n - 1) \neq m \cdot n$$

contradiction.

To complete, we deal with the case of $|U| > m \cdot n$ by considering $U' \subset U$ such that $|U'| = m \cdot n$. From the above, for every function h there is a set $S \subset U' \subset U$, such that h maps all the elements in S to the same cell.